MTH1002

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2022

Mathematical Methods

Module Leader: Dr Tim Jupp

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No word count specified

SECTION A

1. (a) Find the critical point of

$$f(x, y) = x^2 - 4xy + 2y - y^2,$$

and determine whether it is a local minimum, a local maximum, or a saddle point. (10)

(b) Express the definite integral

$$\int_{-1}^{3} (2x - 5)\, dx$$

as a Riemann sum and hence evaluate it. Verify your answer by calculating the integral using the rules of integration. (5)

(c) Evaluate the following definite integrals by using suitable substitutions:

(i)

$$\int_0^1 \frac{t\, dt}{\sqrt[3]{1 + 7t^2}},$$

(ii)

$$\int_0^{3/2} x^2 \sqrt{9 - x^2}\, dx.$$

(10)

(d) Using Gaussian elimination, find the inverse (if it exists) of

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 1 & 0 \\ 1 & -2 & 1 \end{pmatrix}.$$

(10)

(e) Solve the ordinary differential equations:

(i)

$$\frac{dy}{dx} + \frac{2}{x} y = 1 + x^3 \quad \text{with } y(1) = 1,$$

(ii)

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = \sin x \quad \text{with } y(0) = \frac{-3}{10},\ y'(0) = \frac{11}{10}.$$

(10)

(f) Suppose that $f(x) = 2x^2 - 3x + 2$ and consider

$$A = \begin{pmatrix} 2 & 1 \\ 0 & -3 \end{pmatrix}.$$

Find the matrix $f(A)$. (5)

[**50**]

SECTION B

2. (a) Using Elementary Row Operations, or otherwise, evaluate the determinants of the following matrices.

(i)
$$\begin{pmatrix} 1 & 3 & -2 & 1 \\ 2 & 8 & -1 & 6 \\ -1 & 3 & 7 & 3 \\ 1 & 3 & 0 & 8 \end{pmatrix};$$

(ii)
$$\begin{pmatrix} 1 & ab & c \\ 1 & bc & a \\ 1 & ca & b \end{pmatrix}.$$

(For full marks in part (ii) you should give the determinant of the 3-by-3 matrix as a product of three factors.) (10)

(b) Determine whether each of the following statements about $n \times n$ matrices A and B is always true or can be false. Justify your answers by giving a proof or a counterexample.

(i) $(A+B)^2 = A^2 + 2AB + B^2$;

(ii) if $A^2 = I$ then A is invertible;

(iii) if $AB = B$ then B is invertible;

(iv) $(AB)^T = A^T B^T$;

(v) $(AA^T)^T = AA^T$.

(Here, I denotes the $n \times n$ identity matrix, and A^T denotes the transpose of A.) (15)

[**25**]

3. (a) Suppose $x > 0$, $y > 0$ and $x + y = 2c$ where c is constant. By examining the critical point of a suitable function, show that

$$\frac{1}{2}\left(x^{11} + y^{11}\right) \geq \left(\frac{x+y}{2}\right)^{11}.$$

(10)

(b) Let $z(u)$ be a twice–differentiable function of a single variable u that is itself a function of independent variables x and y:

$$u(x, y) = \sqrt{x^2 + y^2}.$$

Suppose that z (when regarded as a function of x and y) satisfies the partial differential equation

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = x^2 + y^2.$$

Find the function z.

Hint: Obtain an ordinary differential equation for $z(u)$ and then solve it. (15)

[**25**]

4. (a) Evaluate the integral

$$\int_{-1}^{2} \int_{y^2-2y}^{2-y} (x+y)\ dx\ dy.$$

(8)

(b) Sketch the region of the plane defined by the limits of integration of the previous integral. (8)

(c) Give the form of the area element in the (x, y) plane for polar coordinates (r, θ) defined in the standard way. Justify your answer (for example by a sketch).

Evaluate the integral

$$\int_D x^2 y^2\ dx\ dy,$$

where D is a disc of radius a, centre the origin, in the (x, y)-plane. (9)

[**25**]